

Objective

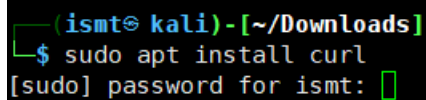
Splunk configuration to understand basic SIEM function

Linux Server

1. Keep Kali Linux system updated.

```
$ sudo apt -y update  
$ sudo apt -y upgrade
```

2. Install Curl Package in Kali Linux which is required by splunk



```
(ismt@kali)-[~/Downloads]  
└─$ sudo apt install curl  
[sudo] password for ismt: 
```

3. Download Trail version of Splunk Enterprise

From Official Website:

<https://www.splunk.com/>

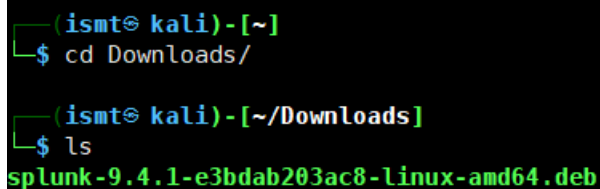
Or,

Google Drive download link:

https://drive.google.com/drive/folders/1h9ZevccIrfITuMVxGy94m2pIVJ8dwCva?usp=drive_link

4. Copy downloaded Splunk package to Downloads folder of Kali Linux. Verify it using following commands

```
$ cd Downloads/  
$ ls
```



```
(ismt@kali)-[~]  
└─$ cd Downloads/  
  
(ismt@kali)-[~/Downloads]  
└─$ ls  
splunk-9.4.1-e3bdab203ac8-linux-amd64.deb
```

5. Install the splunk package in Kali Linux

```
(ismt@kali)-[~/Downloads]
$ sudo dpkg -i splunk-9.4.1-e3bdab203ac8-linux-amd64.deb
[sudo] password for ismt: 
```

6. After installation is complete, use following command to accept the license agreement.

\$ sudo /opt/splunk/bin/splunk start --accept-license --answer-yes

```
(ismt@kali)-[~/Downloads]
$ sudo /opt/splunk/bin/splunk start --accept-license --answer-yes
```

Above command will prompt for administrative username and password setup

```
(ismt@kali)-[~/Downloads]
$ sudo /opt/splunk/bin/splunk start --accept-license --answer-yes

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: 
```

For this lab, following username and password is provided:

Username: ismt

Password: Ismt@12345

Once the setup is completed, login URL is also displayed.

Login URL: http://127.0.0.1:8000

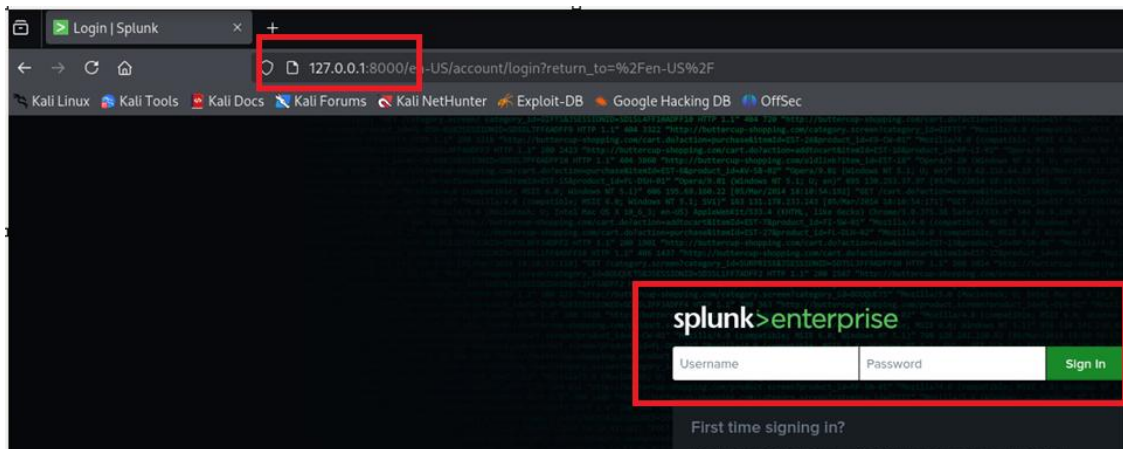
```
Waiting for web server at http://127.0.0.1:8000 to be available..... Done
```

7. Start splunk service to access the URL

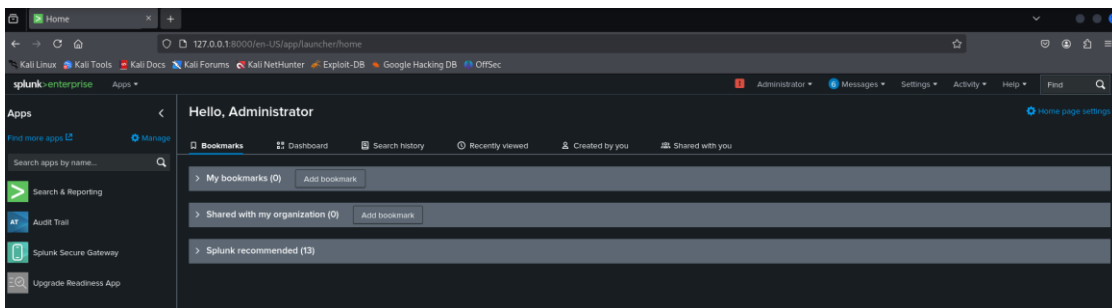
\$ sudo /opt/splunk/bin/splunk start

```
(ismt@kali)-[~/Downloads]
$ sudo /opt/splunk/bin/splunk start
```

8. Check if the Splunk web URL is accessible via Browser in Kali Linux

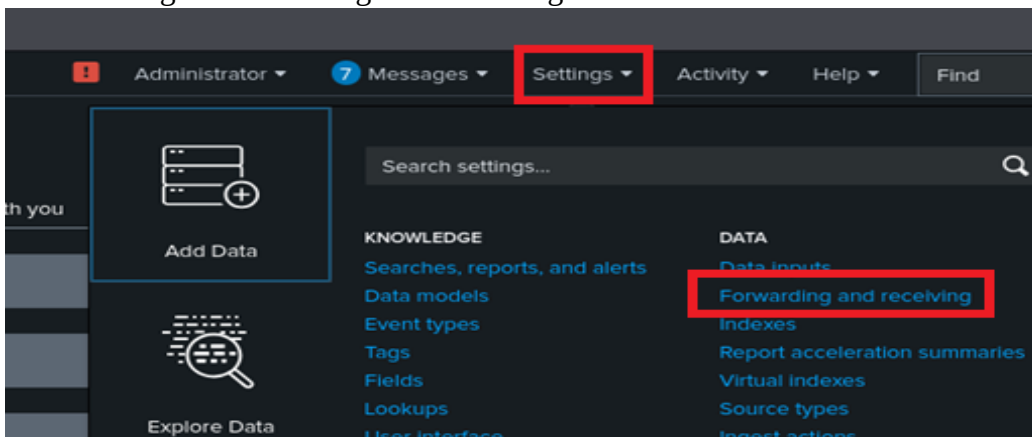


9. Login to the Web Portal using Username and Password provided in Step No – 6.



10. Since Splunk application is a SIEM solution, it should be configured to receive/accept logs from other assets.

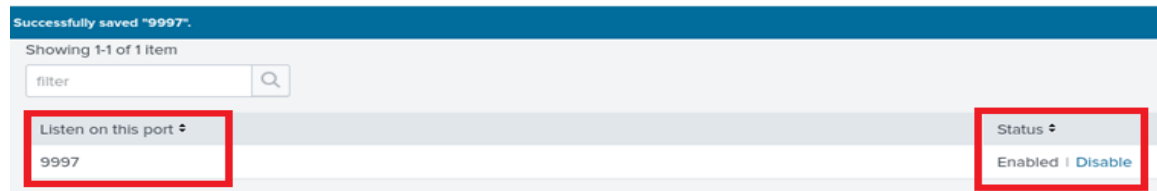
Go to Settings > Forwarding and Receiving



Under receive data section click on Add New



Provide Listening Port as “9997” and save it



11. We are configuring splunk in virtual environment where the storage is low, hence we need to make some adjustment in splunk configuration file.

```
$ cd /opt/splunk/etc/system/local
$ sudo cp server.conf server.conf.bkp
```

```
(ismt@kali)-[/opt/splunk/etc/system/local]
└─$ cd /opt/splunk/etc/system/local

(ismt@kali)-[/opt/splunk/etc/system/local]
└─$ sudo cp server.conf server.conf.bkp
```

```
$ sudo vi server.conf
```

Add the following at the end of the configuration file and save it.

```
[diskUsage]
minFreeSpace = 50
```

```
[general]
serverName = kali
pass4SymmKey = $7$EB2Jv1wk8+Xjn522j26b3ftuPuadUe9+vCUU+RGa3yoqv+4DY/L+mA==

[sslConfig]
sslPassword = $7$YRX3sBF0Th15uPzTJgGF8VyDRlrrjodDY2TXmGbvvksGT1IJSWIKCA==

[lmppool:auto_generated_pool_download-trial]
description = auto_generated_pool_download-trial
peers = *
quota = MAX
stack_id = download-trial

[lmppool:auto_generated_pool_forwarder]
description = auto_generated_pool_forwarder
peers = *
quota = MAX
stack_id = forwarder

[lmppool:auto_generated_pool_free]
description = auto_generated_pool_free
peers = *
quota = MAX
stack_id = free

[diskUsage]
minFreeSpace = 50
```

Verify if the configuration is properly saved.

\$ sudo cat server.conf

12. Stop and Start the service for configuration to reload

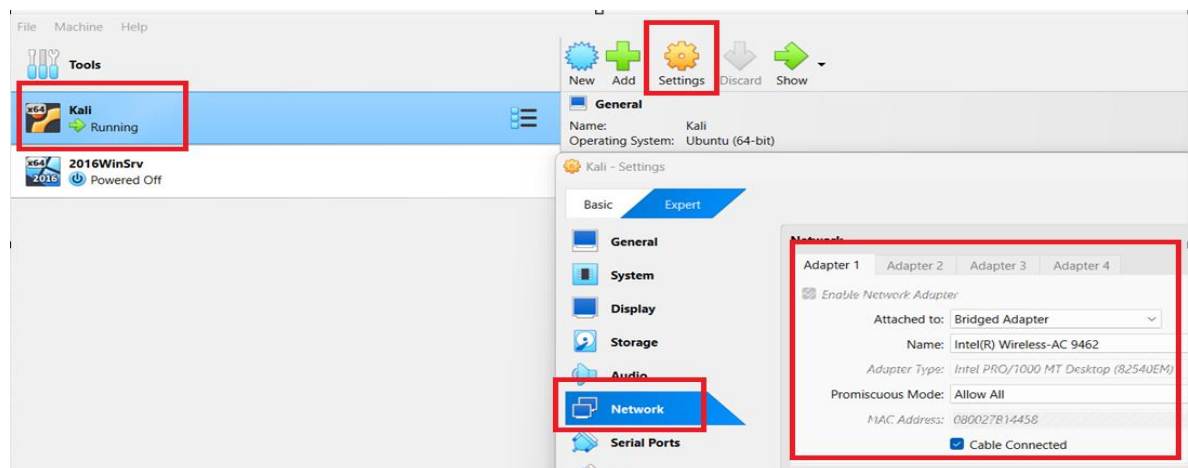
\$ sudo /opt/splunk/bin/splunk stop

\$ sudo /opt/splunk/bin/splunk start

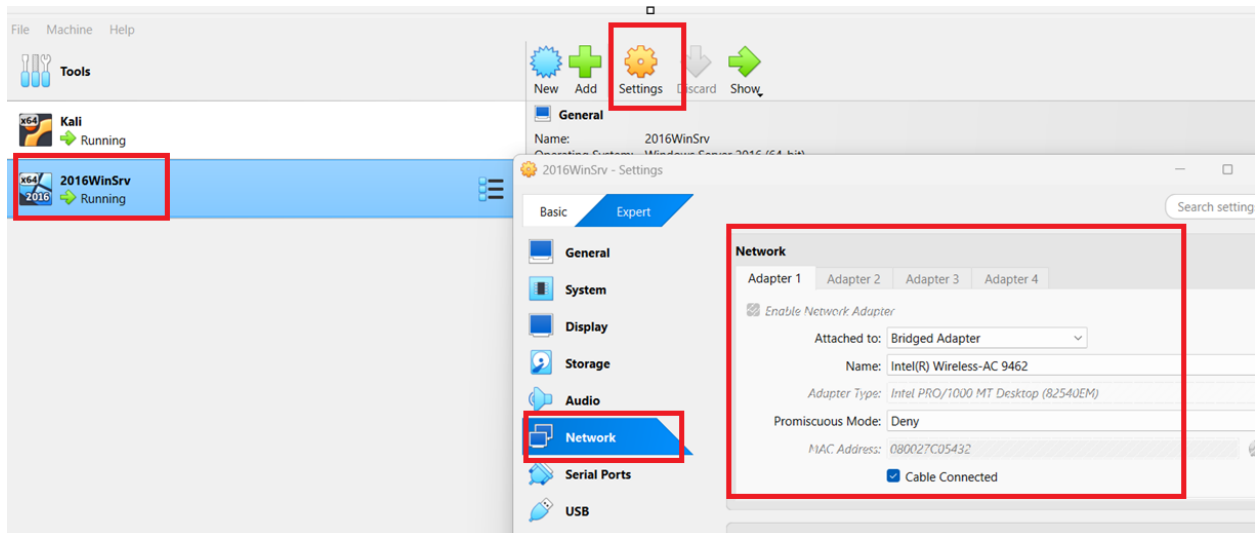
Virtual Box

For this Lab Purpose let us configure different Virtual Box Network for smooth operation.

1. Select Kali Linux Box > Click on Setting > Network

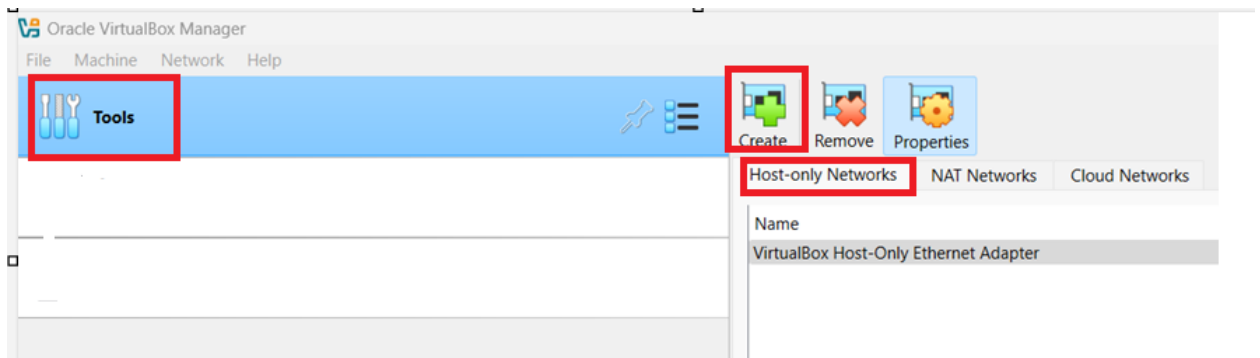


Select Windows Server Box > Click on Settings > Network



Note: Please keep this screenshot safe as we have to revert back the setting to this state after completion of this Lab.

2. Go to Tools > Select Host-Only Networks and Click on Create



New VirtualBox Host-Only Adaptor will be created.



Change the Adapter IP address. For this LAB, “**10.10.35.0/24**” is used.

The screenshot shows the 'Adapter' tab of a configuration window. The 'Configure Adapter Manually' option is selected and highlighted with a red box. Below it, the 'IPv4 Address' is set to '10.10.35.0' and the 'IPv4 Network Mask' is set to '255.255.255.0', both also highlighted with a red box. The 'IPv6 Address' is 'fe80::9420:c2f3:65b9:5263' and the 'IPv6 Prefix Length' is '64'.

Enable DHCP server for this adaptor and set up DHCP parameters as shown

The screenshot shows the 'DHCP Server' tab. The 'Enable Server' checkbox is checked and highlighted with a red box. Below it, the 'Server Address' is '10.10.35.2', 'Server Mask' is '255.255.255.0', 'Lower Address Bound' is '10.10.35.3', and 'Upper Address Bound' is '10.10.35.100'. These four fields are grouped together and highlighted with a red box.

Click on Apply

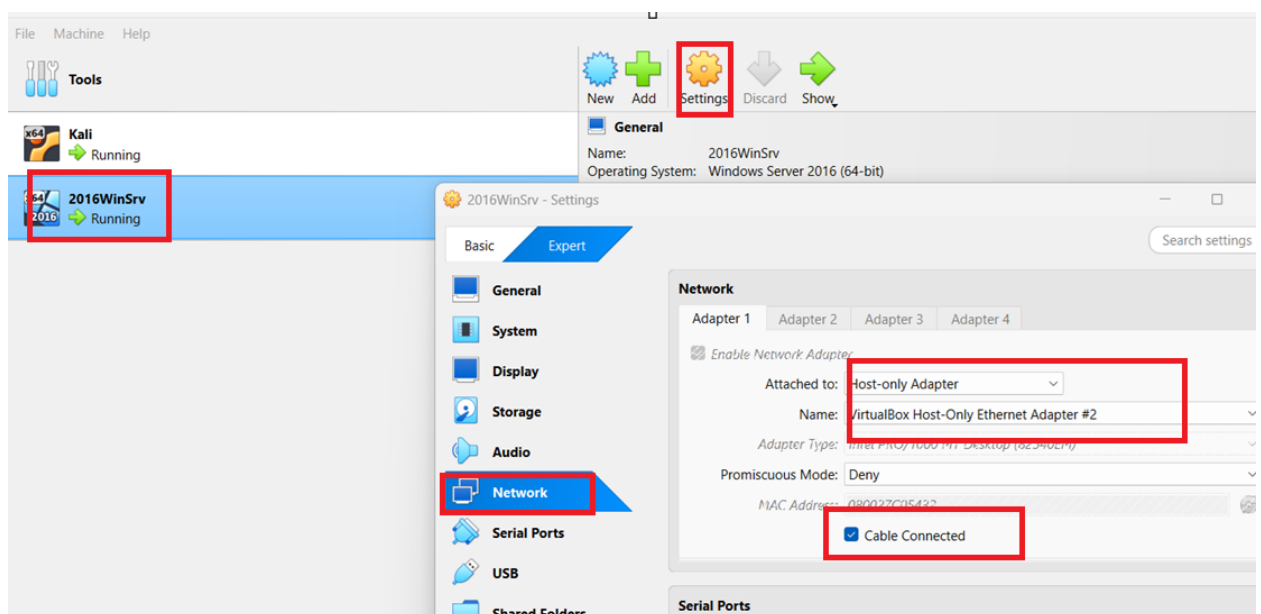
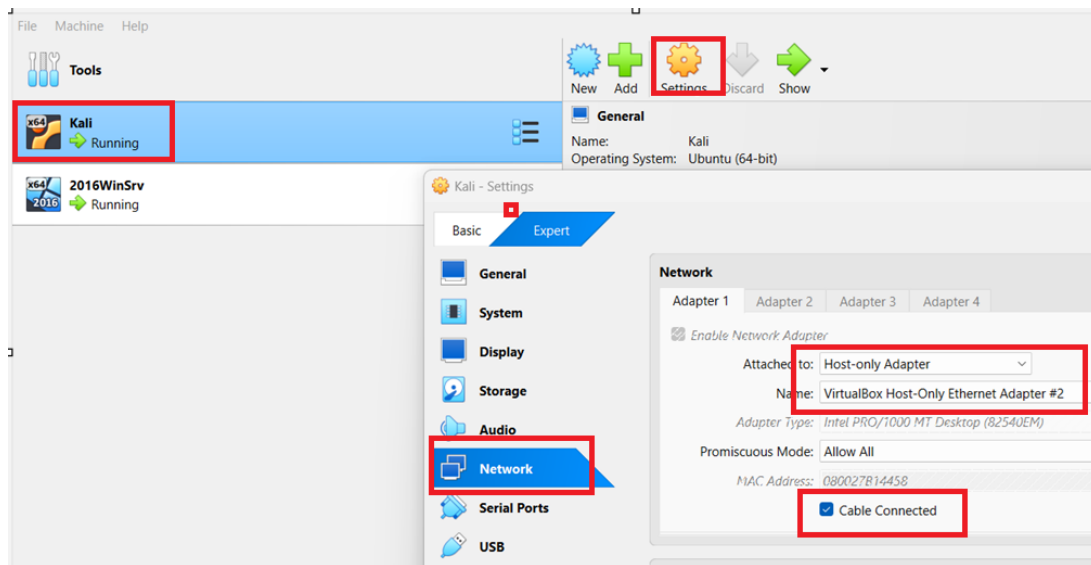
This screenshot shows the same configuration window as before, but with the 'Apply' button at the bottom right highlighted with a red box. The 'Reset' button is also visible next to it.

Finally, New VirtualBox Host Only Adaptor will be created.

The screenshot shows the 'Host-only Networks' tab in VirtualBox. A table lists the networks. The first row, 'VirtualBox Host-Only Ethernet Adapter #2', is highlighted with a red box. It has an IPv4 Prefix of '10.10.35.0/24' and its DHCP Server is 'Enabled'. The second row, 'VirtualBox Host-Only Ethernet Adapter', has an IPv4 Prefix of '192.168.56.1/24' and its DHCP Server is also 'Enabled'.

Name	IPv4 Prefix	IPv6 Prefix	DHCP Server
VirtualBox Host-Only Ethernet Adapter #2	10.10.35.0/24		Enabled
VirtualBox Host-Only Ethernet Adapter	192.168.56.1/24		Enabled

3. Assign this Adaptor to Kali Linux and Windows Server.



4. Check the IP Address received by Kali Linux and Windows Server.




```
C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 10.10.35.4
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Tunnel adapter isatap.{FA18D936-946E-44E4-9F2D-19D3A0466497}:
```

In this Lab, Following IP Address are assigned via DHCP.

Kali Linux – 10.10.35.3

Window Server – 10.10.35.4

Note: These IP are essential for this LAB, please make note of these IPs.

Windows Server

1. Download Splunk Universal Forwarder

Form Official Website:

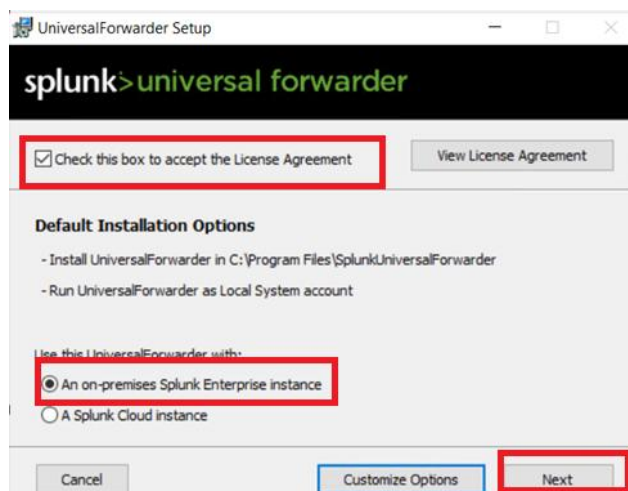
<https://www.splunk.com/>

Or,

From Google Drive Link:

https://drive.google.com/drive/folders/1h9ZevccIrfITuMVxGy94m2pIVJ8dwCva?usp=drive_link

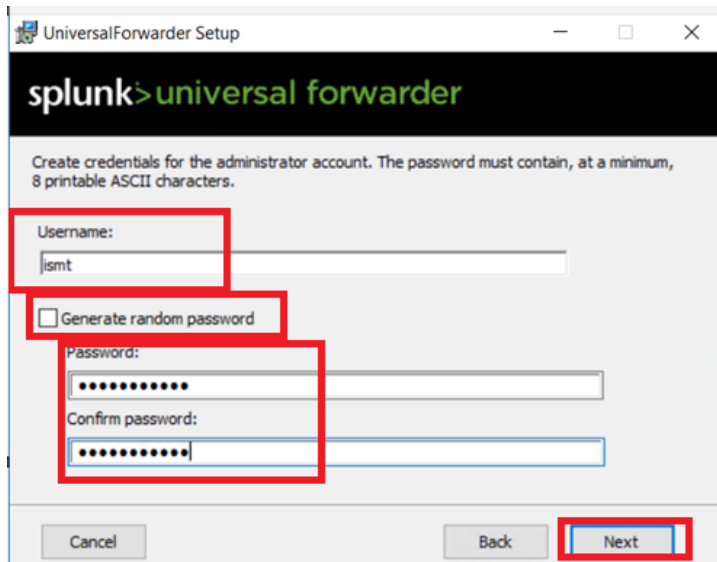
2. Double Click Splunk Universal Forwarder in Windows Server.



3. Prompt to create administrative user will be provided. For this Lab following details provided.

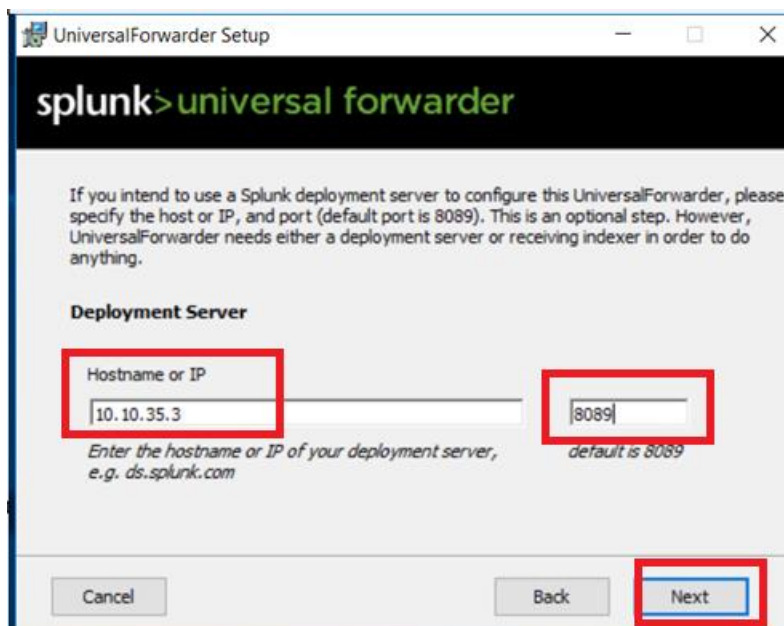
Username: ismt

Password: Ismtlab@123



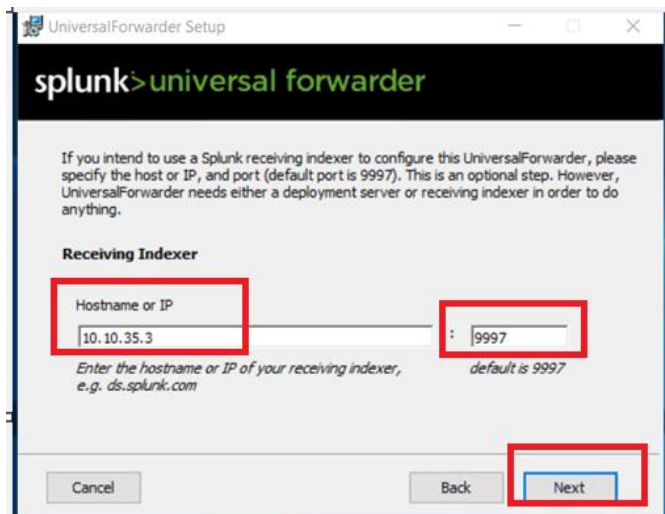
The screenshot shows the 'UniversalForwarder Setup' window. The title bar says 'UniversalForwarder Setup'. The main header has the Splunk logo and 'universal forwarder'. Below the header, it says 'Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters.' There are three input fields: 'Username:' with 'ismt' entered, 'Password:' with 'Ismtlab@123' entered (masked with dots), and 'Confirm password:' with 'Ismtlab@123' entered (masked with dots). There is a checkbox labeled 'Generate random password' which is unchecked. At the bottom, there are three buttons: 'Cancel', 'Back', and 'Next'. The 'Next' button is highlighted with a red box.

4. In Next step, IP Address of Splunk server, which is Kali Linux in this lab, is to be provided. As identified in step 4, IP address – 10.10.35.3 is provided and default port 8089 is provided.

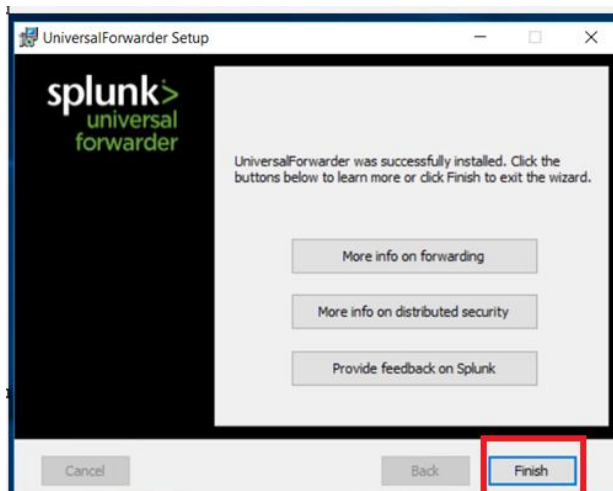


The screenshot shows the 'UniversalForwarder Setup' window. The title bar says 'UniversalForwarder Setup'. The main header has the Splunk logo and 'universal forwarder'. Below the header, it says 'If you intend to use a Splunk deployment server to configure this UniversalForwarder, please specify the host or IP, and port (default port is 8089). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.' There is a section titled 'Deployment Server' with two input fields: 'Hostname or IP' with '10.10.35.3' entered, and 'Port' with '8089' entered. Below the 'Hostname or IP' field, it says 'Enter the hostname or IP of your deployment server, e.g. ds.splunk.com'. Below the 'Port' field, it says 'default is 8089'. At the bottom, there are three buttons: 'Cancel', 'Back', and 'Next'. The 'Next' button is highlighted with a red box.

5. In Next step, IP Address of Receiver Indexer, which is Kali Linux in this lab, is to be provided. As identified in step 4, IP address – 10.10.35.3 is provided and default port 9997 is provided.



6. Click on Next until installation is completed.



Kali Linux

Now, we need to add this windows server in Splunk for accepting logs.

1. Check the status of Splunk if it is running or not.

```
$ sudo /opt/splunk/bin/splunk status
```

```
(ismt@ kali)-[~]  
$ sudo /opt/splunk/bin/splunk status  
splunkd is not running.
```

As shown above, splunk service is not running. Hence it needs to be started. If the service is in running state, no need to execute below command.

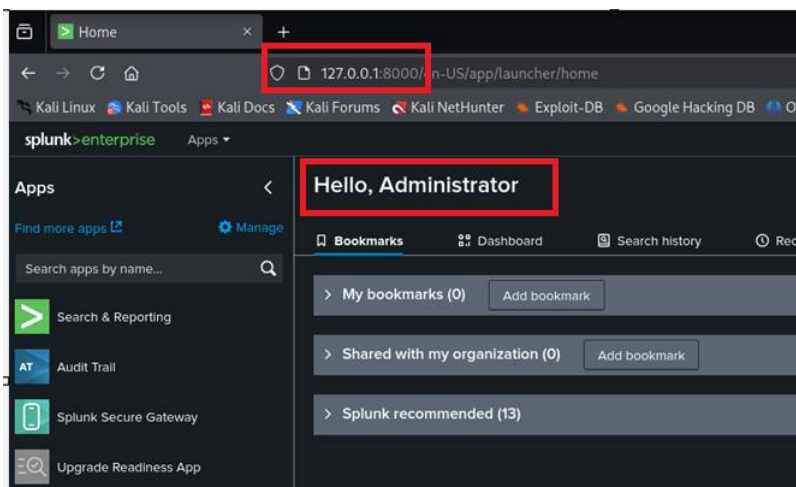
\$ sudo /opt/splunk/bin/splunk start

\$ sudo /opt/splunk/bin/splunk status

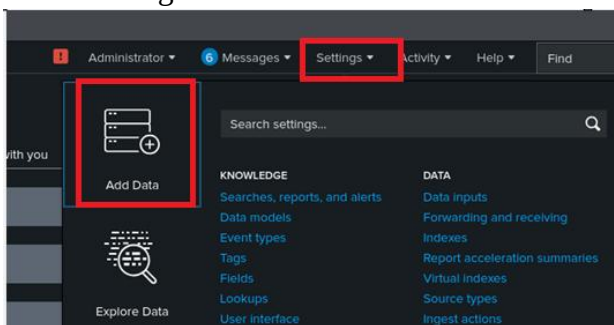
```
(ismt@kali)-[~]  
$ sudo /opt/splunk/bin/splunk status  
splunkd is running (PID: 144790).  
splunk helpers are running (PIDs: 144791 144991 145087 145134 145139 145146).
```

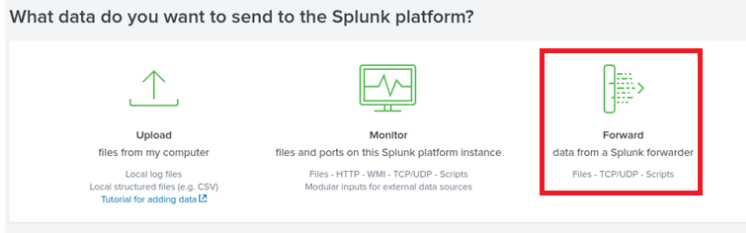
2. Access web portal for splunk, using username and password created above in step 6.

http:// 127.0.0.1:8000



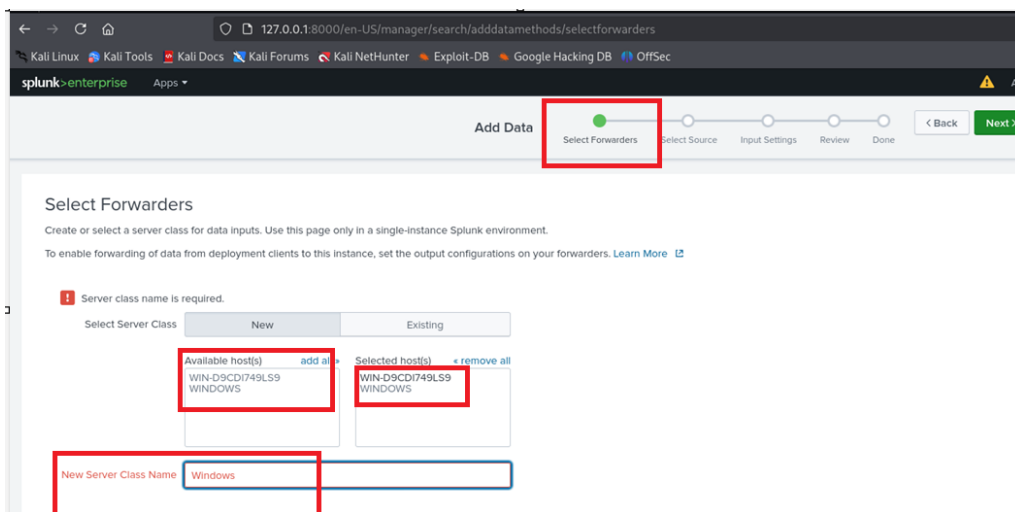
3. Go to Setting > Add Data > Forward



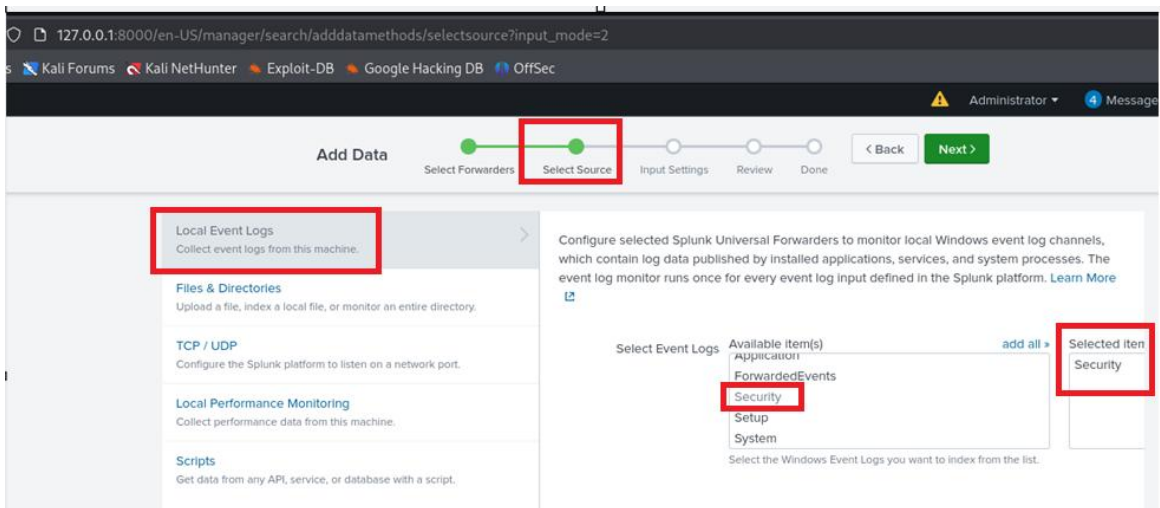


4. Click on available host to select as a forwarder, once selected it should be added to selected host section as well.

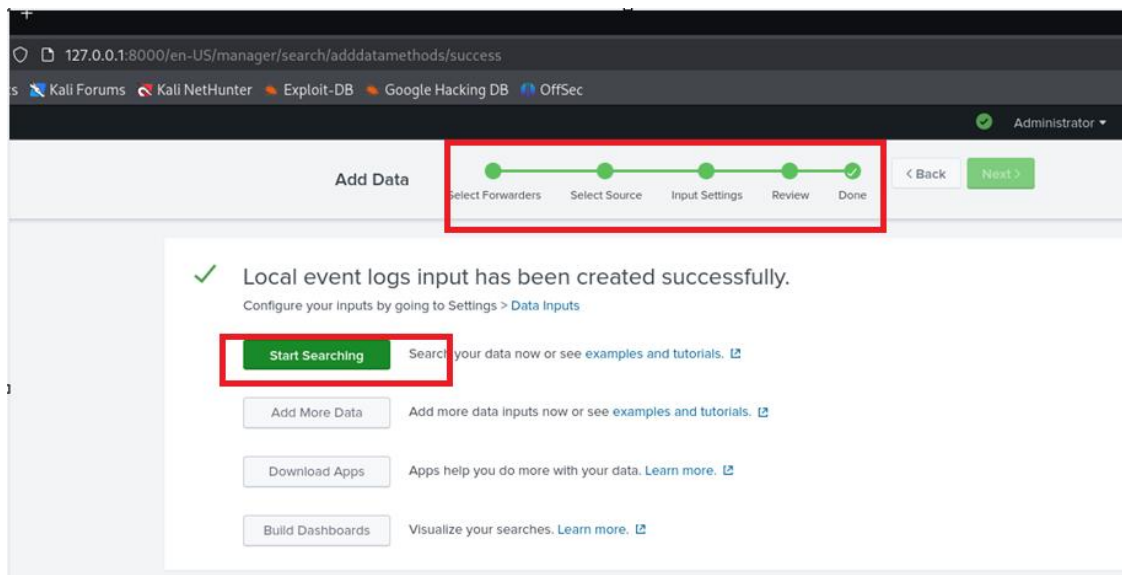
Add New Server Class Name as “Windows”. Finally click on Next



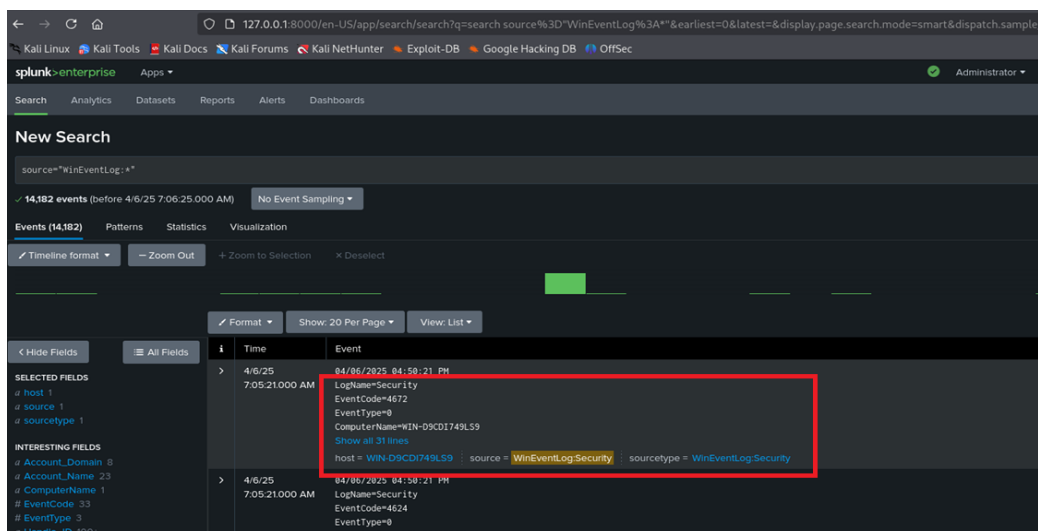
5. In select source option, select Local Event Logs and event type as security only. Then click on next.



- Click next until configuration is completed and start searching for logs.



- Finally we will see Security related event logs from Windows Server.



- Let us test with a special use case.

Windows

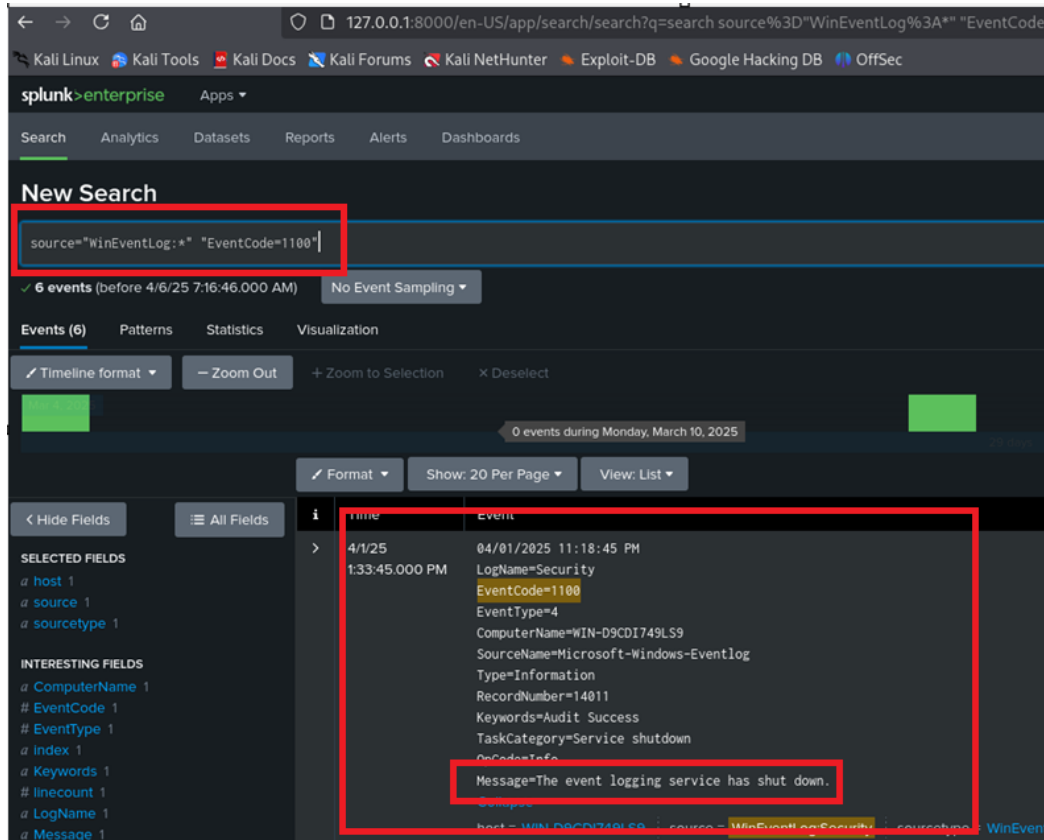
Restart (Reboot) windows server.

Kali Linux (Splunk)

Restarting event is captured with Windows Event Log ID – 1100. If we search for this specific id, information related to windows restart will be searched.

On Search Panel, search for:

source="WinEventLog:*" "EventCode=1100"



After Lab is completed, stop splunk service and revert the VM Network Adaptor to previous state as per screenshots in Virtual Box Step 1.

\$ sudo /opt/splunk/bin/splunk stop

```
(ismt@kali)-[~]
└─$ sudo /opt/splunk/bin/splunk stop
[sudo] password for ismt:
Stopping splunkd...
Shutting down. Please wait, as this may take a few minutes.
....
Stopping splunk helpers...
Done.
```